

Life Processes — MCQ Worksheet

Science • Chapter 5 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Photosynthesis takes place in:

- (A) Mitochondria
- (B) Chloroplasts
- (C) Ribosomes
- (D) Nucleus

Q2. The site of energy production in a cell is:

- (A) Nucleus
- (B) Chloroplast
- (C) Mitochondrion
- (D) Ribosome

Q3. Mode of nutrition in fungi is:

- (A) Holozoic
- (B) Saprophytic
- (C) Parasitic
- (D) Autotrophic

Q4. The opening of leaves through which gases exchange is called:

- (A) Stomata
- (B) Lenticels
- (C) Xylem
- (D) Phloem

Q5. Saliva contains the enzyme:

- (A) Pepsin
- (B) Trypsin
- (C) Salivary amylase
- (D) Lipase

Q6. The function of HCl in the stomach is to:

- (A) Digest fats
- (B) Provide an acidic medium for pepsin
- (C) Absorb water
- (D) Produce bile

Q7. Bile is secreted by the:

- (A) Pancreas
- (B) Liver
- (C) Stomach
- (D) Small intestine

Q8. Most absorption of digested food occurs in:

- (A) Stomach
- (B) Small intestine (villi)
- (C) Large intestine
- (D) Oesophagus

Q9. During heavy exercise, glucose breaks down anaerobically in muscles to give:

- (A) CO₂ and water
- (B) Lactic acid
- (C) Ethanol
- (D) Pyruvate only

Q10. The breakdown of glucose to pyruvate occurs in the:

- (A) Mitochondria
- (B) Cytoplasm
- (C) Nucleus
- (D) Chloroplast

Q11. The structural and functional unit of the kidney is:

- (A) Neuron
- (B) Nephron
- (C) Alveolus
- (D) Hepatocyte

Q12. Filtration of blood occurs in the:

- (A) Henle loop
- (B) Glomerulus inside Bowman's capsule
- (C) Collecting duct
- (D) Ureter

Q13. Haemoglobin is found in:

- (A) WBCs
- (B) Platelets
- (C) RBCs
- (D) Plasma

Q14. The pigment haemoglobin contains the metal:

- (A) Calcium
- (B) Iron
- (C) Magnesium
- (D) Copper

Q15. The chamber that pumps oxygenated blood to the entire body is the:

- (A) Right atrium
- (C) Right ventricle

- (B) Left atrium
- (D) Left ventricle

Q16. Veins carry blood:

- (A) Away from the heart
- (C) Inside organs only

- (B) Towards the heart
- (D) To the lungs only

Q17. Xylem is responsible for transport of:

- (A) Food
- (C) Hormones

- (B) Water and minerals
- (D) Oxygen only

Q18. Translocation of food in plants occurs through:

- (A) Xylem
- (C) Stomata

- (B) Phloem
- (D) Roots only

Q19. In double circulation, blood passes through the heart:

- (A) Once per cycle
- (C) Thrice per cycle

- (B) Twice per cycle
- (D) Not at all

Q20. In humans, gaseous exchange occurs in:

- (A) Bronchi
- (C) Alveoli

- (B) Trachea
- (D) Larynx

Q21. Plants get rid of gaseous wastes mainly through:

- (A) Stomata
- (C) Stem

- (B) Roots
- (D) Flowers

Q22. Dialysis is used when:

- (A) Heart fails
- (C) Liver fails

- (B) Kidneys fail
- (D) Lungs fail

Q23. The energy currency of the cell is:

- (A) DNA
- (C) Glucose

- (B) ATP
- (D) Oxygen

Q24. Nutrition in tapeworm is:

- (A) Saprophytic
- (C) Holozoic

- (B) Parasitic
- (D) Autotrophic

Q25. Photosynthesis equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{---} + 6\text{O}_2$

- (A) C_6H_6
- (C) $\text{C}_6\text{H}_{12}\text{O}_6$

- (B) CH_4
- (D) $\text{C}_2\text{H}_5\text{OH}$